

Level 11: Real-Time Push Notifications & Deep Linking Architecture

React Native Masterclass

Expo SDK 54 / RN 0.81

Production Architecture

11.1 — Push Fundamentals & Dual-Layer Token Architecture

A push notification is a real-time message sent from a backend server to a user's mobile device, allowing the operating system to display an alert banner even when the app is **not actively open**.

What is a Push Token?

A Push Token is **NOT** a permanent physical hardware serial number (like an IMEI or MAC address). It is a cryptographic destination token associated with a **specific app installation** on a device. Tokens can rotate, expire, or become invalid (e.g. app re-installs or OS restores).



Tenglish Explanation:

Push token ante phone permanent IMEI number kaadu. Phone lo aa specific app install ayi unnanthavaraku issue chese oka temporary destination pass. User app delete chesi malli install chesthe kotha token generate avthundi. Backend database lo eppudu latest active token ni store chesukovali.

11.2 — The Backend Trigger Pipeline & Sending Push Payloads

When an event occurs on your backend server (e.g. Swiggy order shipped, WhatsApp message received), the backend looks up the user's active push token and sends an HTTP POST request to the push gateway:

```
// Standard Push Notification JSON Schema
{
  "to": "ExponentPushToken[xxxxxxxxxxxxxxxxxxxxxx]",
  "title": "Order Out for Delivery! 🍕",
  "body": "Your delivery driver Rajesh is 5 minutes away.",
  "sound": "default",
  "badge": 1,
  "data": {
    "screen": "OrderTracking",
    "orderId": "ORD-9821"
  }
}
```

Handling Invalid Tokens (DeviceNotRegistered): When the push gateway returns an error of `DeviceNotRegistered`, the backend database must immediately mark the token as `is_active = false` or delete it.

Tenglish Explanation:

Backend server Expo push endpoint (<https://exp.host/--/api/v2/push/send>) ki request pampisthundi. Andulo `title`, `body`, and deep linking metadata `data: { screen, orderId }` untayi. User app delete chesthe `DeviceNotRegistered` error vasthundi; appudu database lo aa dead token ni deactivate cheyali.

11.3 — Foreground Notification Handling & Handler Configuration

By default, mobile operating systems **suppress** notification alert banners and sounds if the app is already open in the foreground. To force the OS to display the banner, you must configure `setNotificationHandler` :

```
import * as Notifications from 'expo-notifications';

// Configure foreground display behavior
Notifications.setNotificationHandler({
  handleNotification: async () => ({
    shouldShowAlert: true, // Shows popup banner while app is open!
    shouldPlaySound: true, // Plays audio chime!
    shouldSetBadge: false,
  }),
});
```

Tenglish Explanation:

App already open lo unte phone OS default ga notification banner chupinchadu. App open lo unna sare banner ravali ante app root lo `setNotificationHandler` petti `shouldShowAlert: true` cheyali.

11.4 — User Interaction & Tap Response Listeners

There are two distinct states when handling user taps on a notification banner:

App State	Method Used	Description
Backgrounded / Active	<code>addNotificationResponseReceivedListener</code>	Fires when user taps banner while app is sitting in background memory.
Killed / Cold Start	<code>getLastNotificationResponseAsync()</code>	Catches cold-start launch when user taps lockscreen banner to boot the app from scratch.

Tenglish Explanation:

User notification tap chesinappudu background lo unte `addNotificationResponseReceivedListener` catch chesthundi. App completely closed lo unnapudu tap chesi app open chesthe `getLastNotificationResponseAsync()` dwara payload read chestham.

11.5 & 11.6 — Deep Linking Connection & Enterprise Pipeline

By pairing React Navigation's `createNavigationContainerRef()` with notification tap listeners, you can route users directly to the targeted screen with full parameters:

```

import React, { useEffect } from 'react';
import { NavigationContainer, createNavigationContainerRef } from '@react-navigation/native';
import * as Notifications from 'expo-notifications';

export const navigationRef = createNavigationContainerRef();

export const usePushNotifications = () => {
  useEffect(() => {
    // 1. Cold-Start Tap Catch
    Notifications.getLastNotificationResponseAsync().then((res) => {
      const data = res?.notification?.request?.content?.data;
      if (data?.screen && navigationRef.isReady()) {
        navigationRef.navigate(data.screen, data.params);
      }
    });

    // 2. Real-Time Background Tap Listener
    const sub = Notifications.addNotificationResponseReceivedListener((res) => {
      const data = res.notification.request.content.data;
      if (data?.screen && navigationRef.isReady()) {
        navigationRef.navigate(data.screen, data.params);
      }
    });

    return () => sub.remove();
  }, []);
};

```

Tenglish Summary:

Push notification single tap tho user direct ga specific order/chat screen loki velladaniki

`navigationRef.navigate(data.screen, data.params)` vaadatham. Idi complete 6-step enterprise push notification architecture!